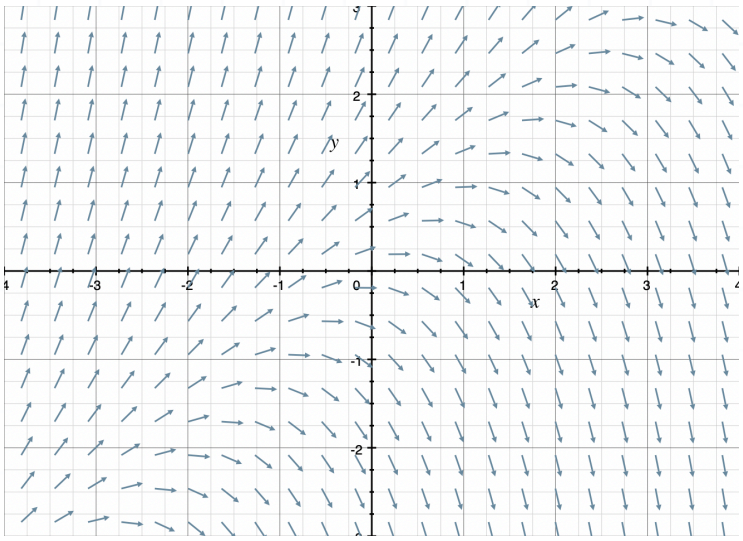
Topic: Direction fields and solution curves

Question: Sketch the direction field of the differential equation.

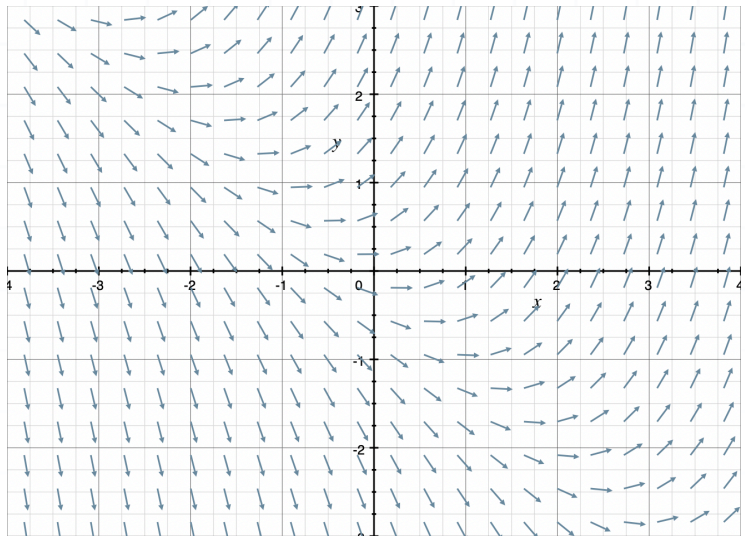
$$y' = y + x$$

Answer choices:

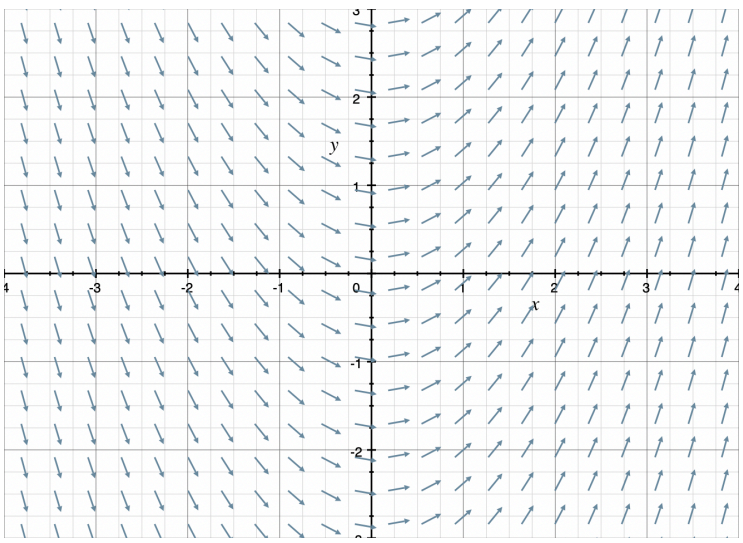
A



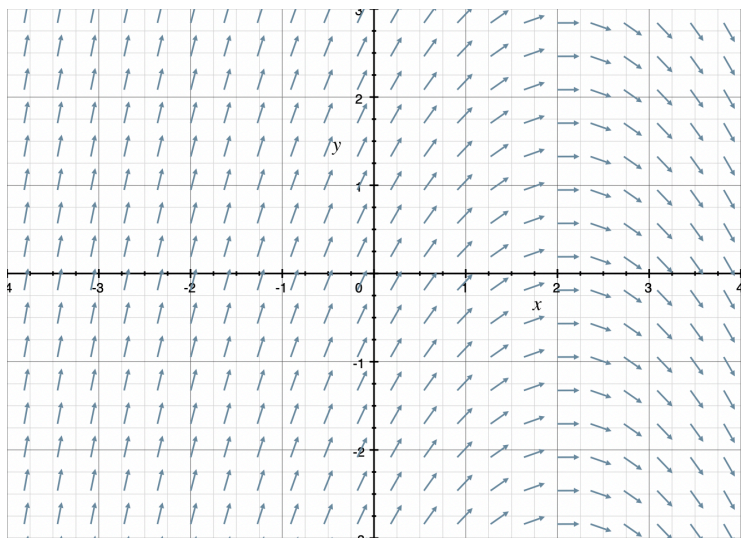
B



C



D



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Solution: B

We'll pair up values $x = \{-2, -1, 0, 1, 2\}$ with values $y = \{-2, -1, 0, 1, 2\}$, calculating y' for each combination of values.

For instance, at $(x, y) = (-2, -2)$, we find

$$y' = y + x$$

$$y' = -2 - 2$$

$$y' = -4$$

Or at $(x, y) = (1, 2)$, we find

$$y' = y + x$$

$$y' = 2 + 1$$

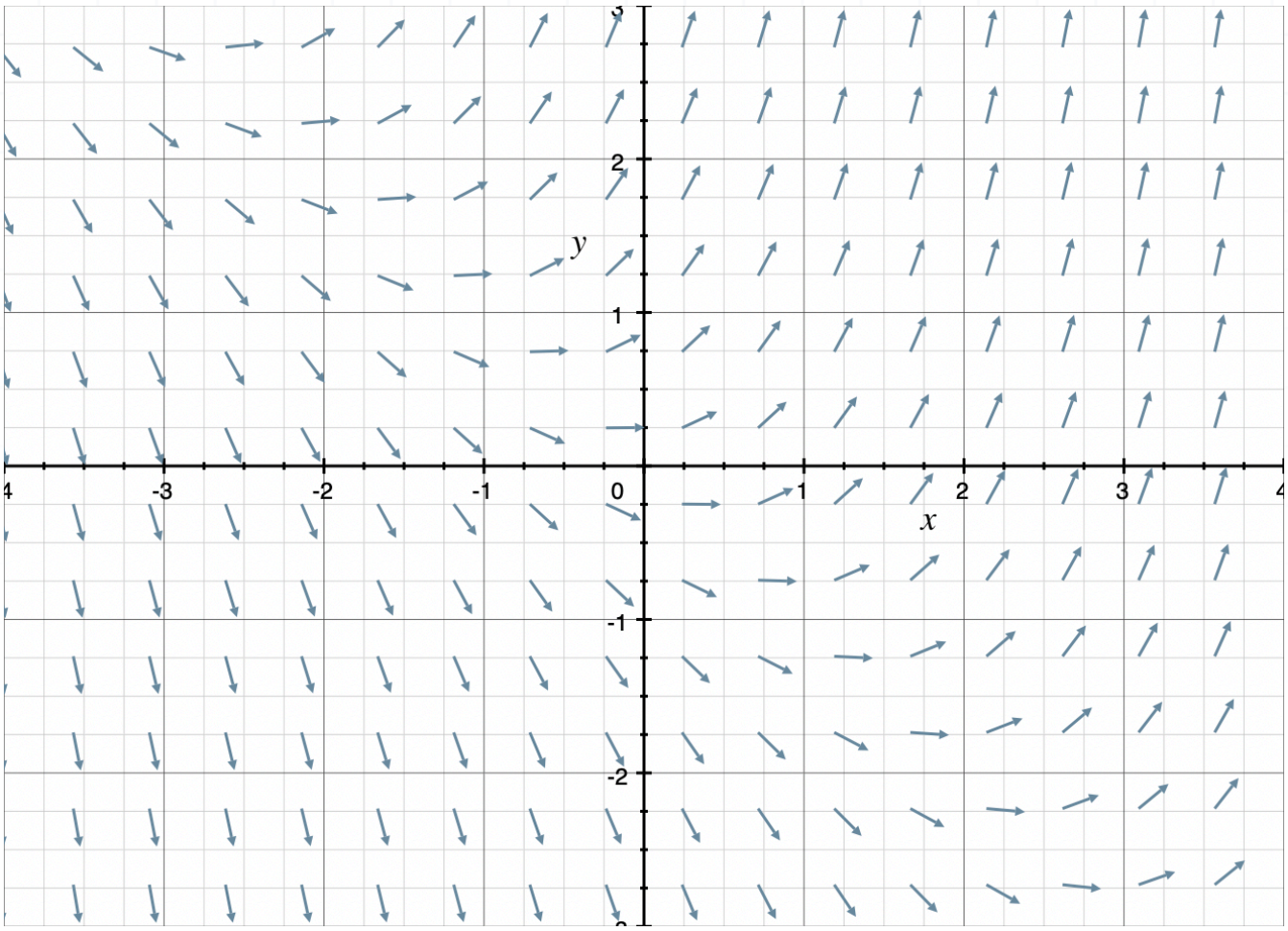
$$y' = 3$$

Let's make a table with each of these combinations.

		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	-4	-3	-2	-1	0
	-1	-3	-2	-1	0	1
	0	-2	-1	0	1	2
	1	-1	0	1	2	3
	2	0	1	2	3	4

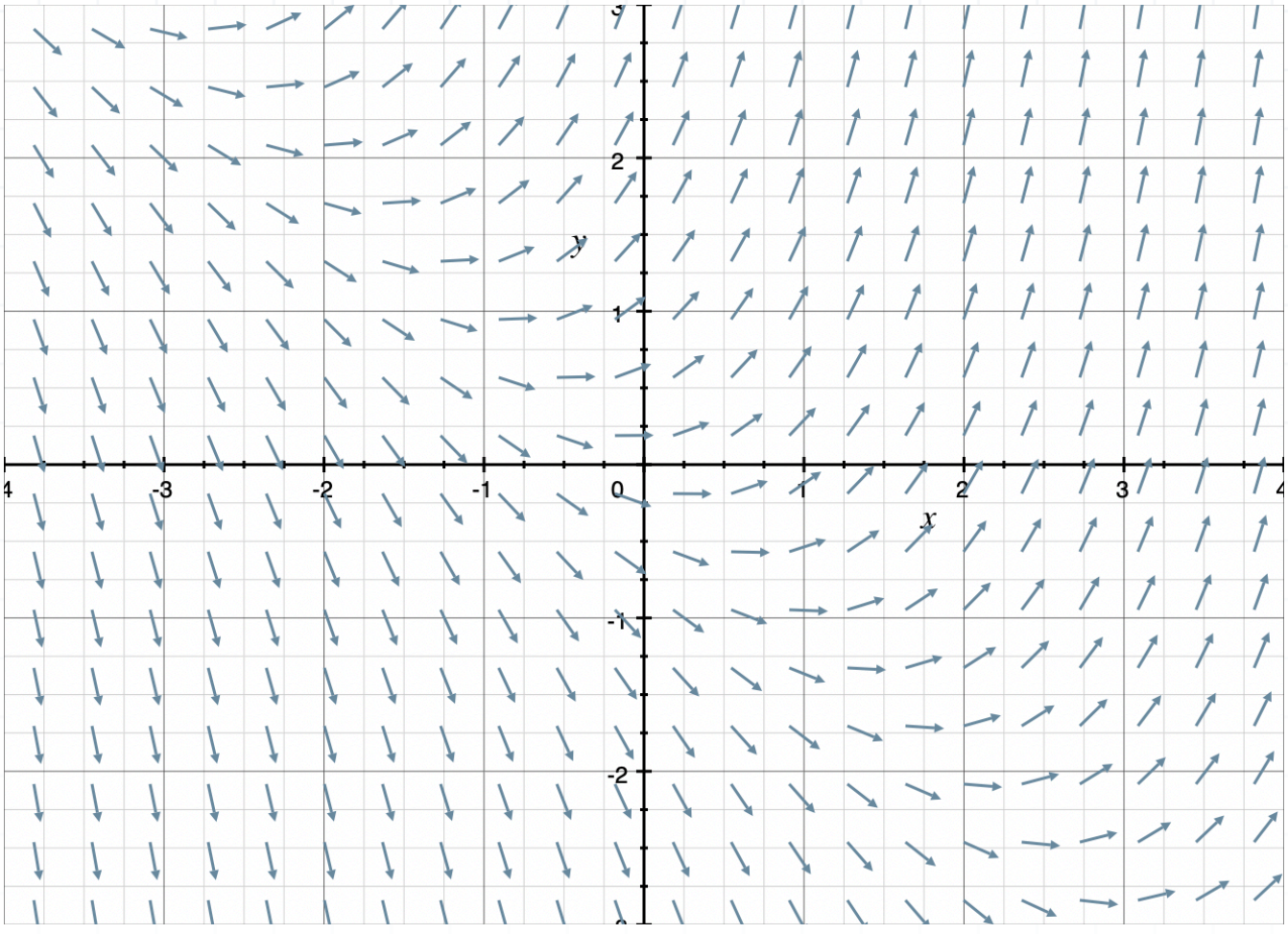
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The values of y' that we found represent the slope of the function at the corresponding point (x, y). For example, we see the point $(2, -2)$, and the corresponding value $y' = 0$. This means that the slope of the function at $(2, -2)$ is 0, so we'd draw a small, short horizontal line exactly at $(2, -2)$. Plotting all of the other point-slope pairs, the direction field starts to look something like this:



If we add more points, maybe ones that are half-way between those that we already found, a more complete direction field should look something like this:

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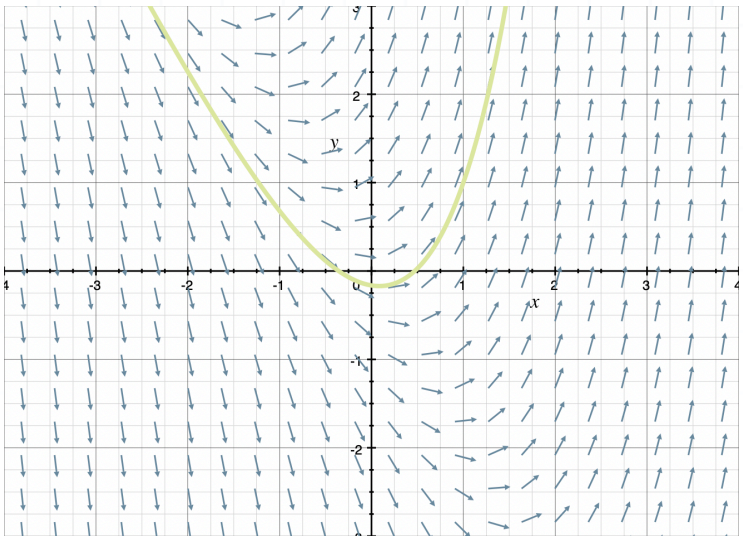
Topic: Direction fields and solution curves

Question: Sketch the direction field for the differential equation, and the solution curve passing through $(1,1)$.

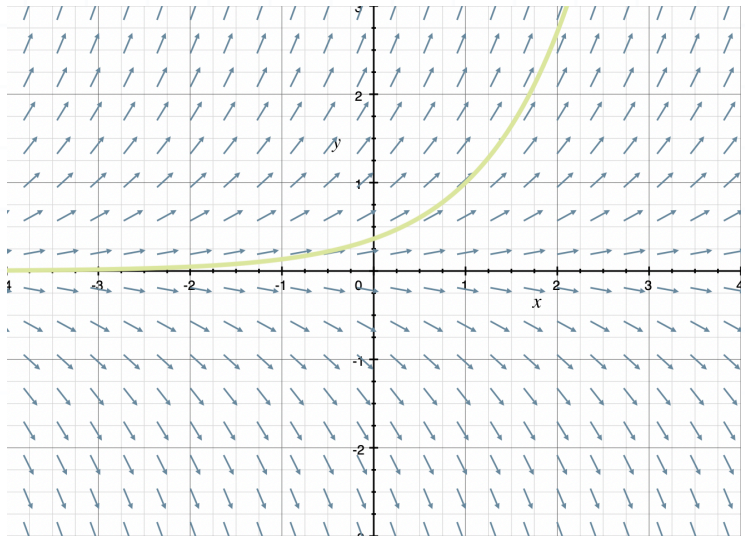
$$y' = y - 2x$$

Answer choices:

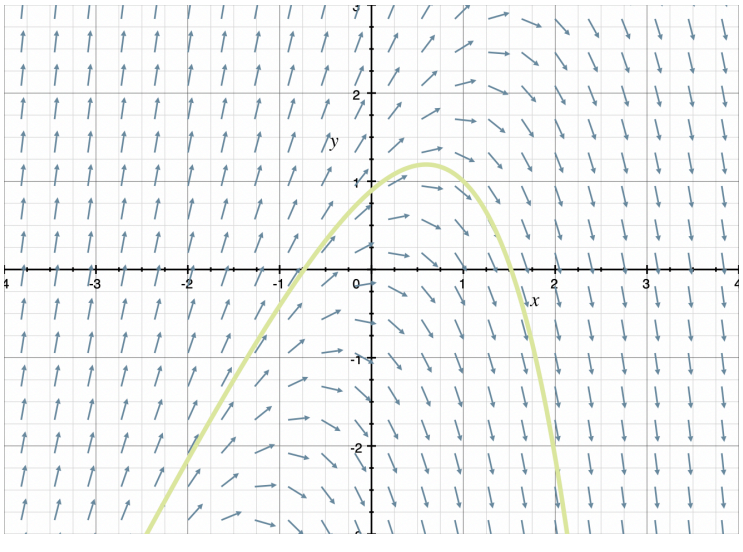
A



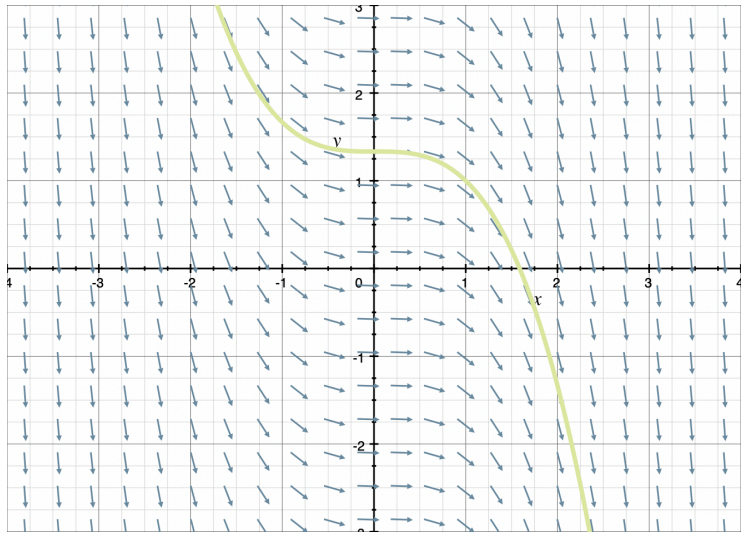
B



C



D



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Solution: C

We'll pair up values $x = \{-2, -1, 0, 1, 2\}$ with values $y = \{-2, -1, 0, 1, 2\}$, calculating y' for each combination of values.

For instance, at $(x, y) = (-2, -2)$, we find

$$y' = y - 2x$$

$$y' = -2 - 2(-2)$$

$$y' = 2$$

Or at $(x, y) = (1, 2)$, we find

$$y' = y - 2x$$

$$y' = 2 - 2(1)$$

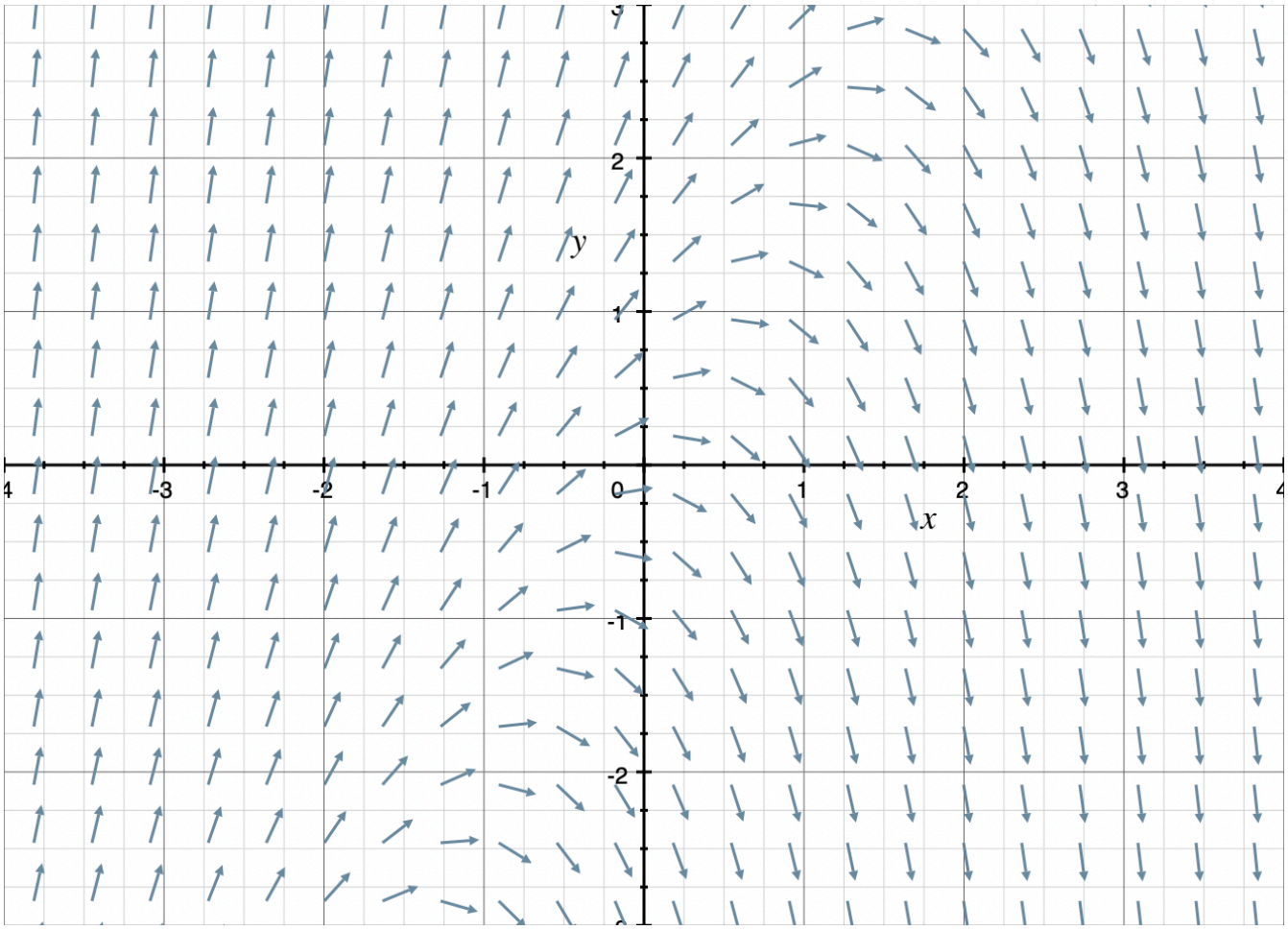
$$y' = 0$$

Let's make a table with each of these combinations.

		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	2	0	-2	-4	-6
	-1	3	1	-1	-3	-5
	0	4	2	0	-2	-4
	1	5	3	1	-1	-3
	2	6	4	2	0	-2

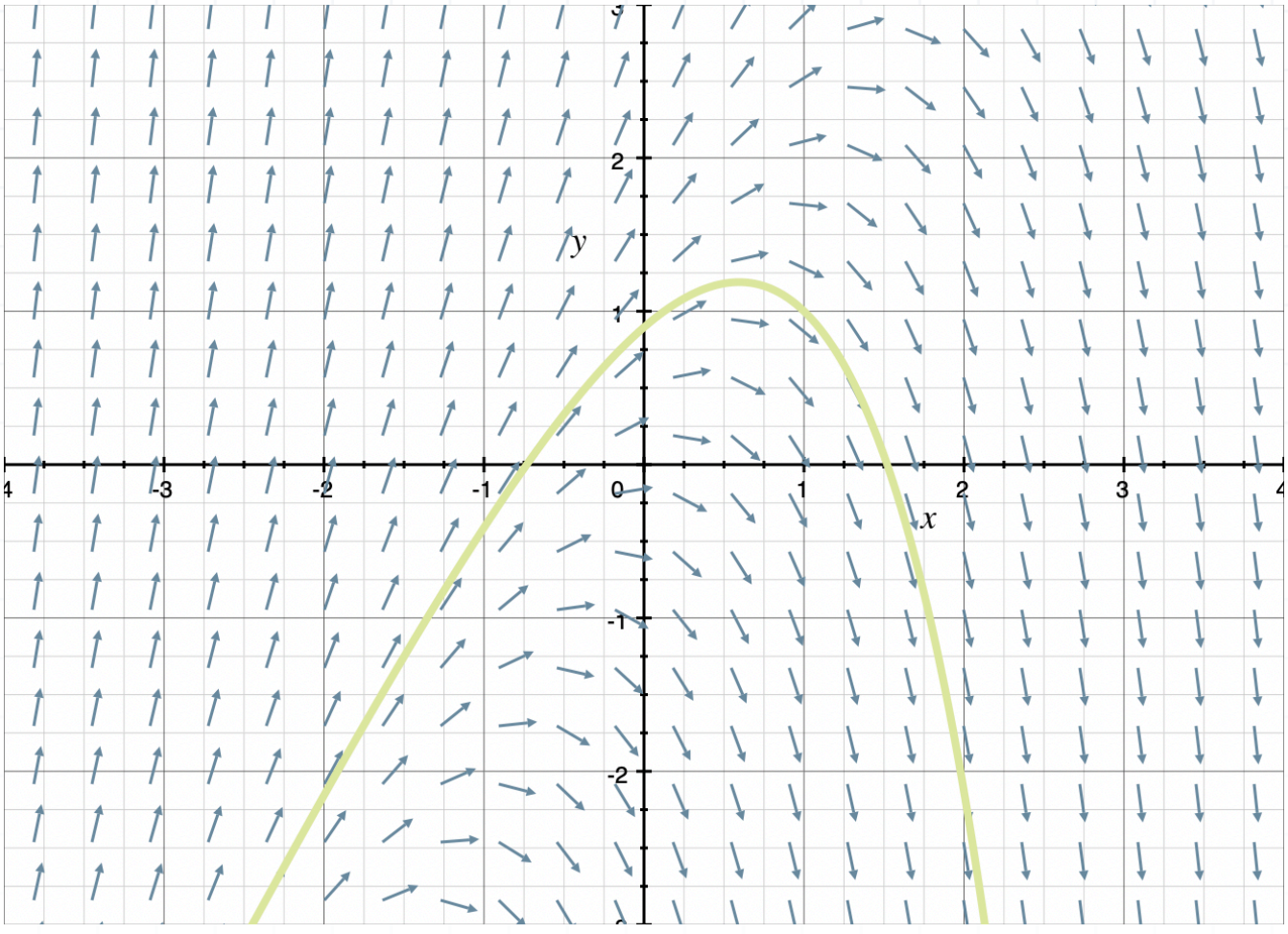
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The values of y' that we found represent the slope of the function at the corresponding point (x, y). For example, we see the point $(2, -2)$, and the corresponding value $y' = -6$. This means that the slope of the function at $(2, -2)$ is -6, so we'd draw a small, short line exactly at $(2, -2)$ with slope -6. Plotting all of the other point-slope pairs, the direction field starts to look something like this:



If we plot the point $(1,1)$, then follow the slopes in our direction field to both the left and right of $(1,1)$, our solution curve should look something like this:

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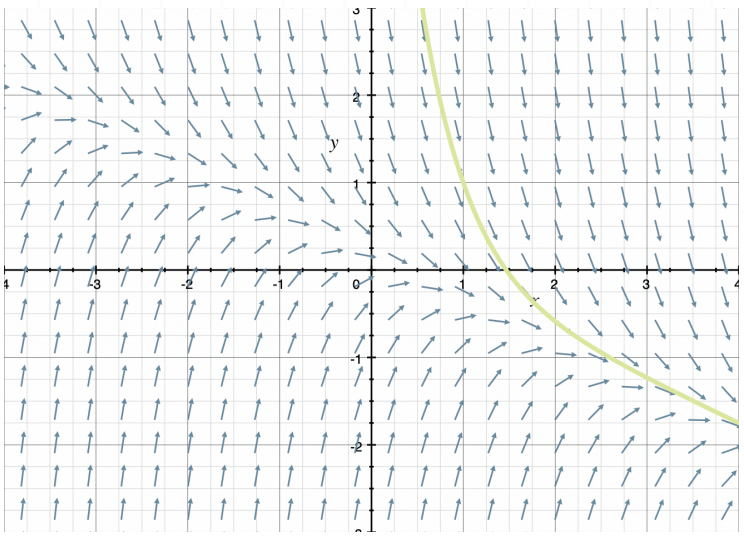
Topic: Direction fields and solution curves

Question: Sketch the direction field for the differential equation and the solution curve passing through $(1,1)$.

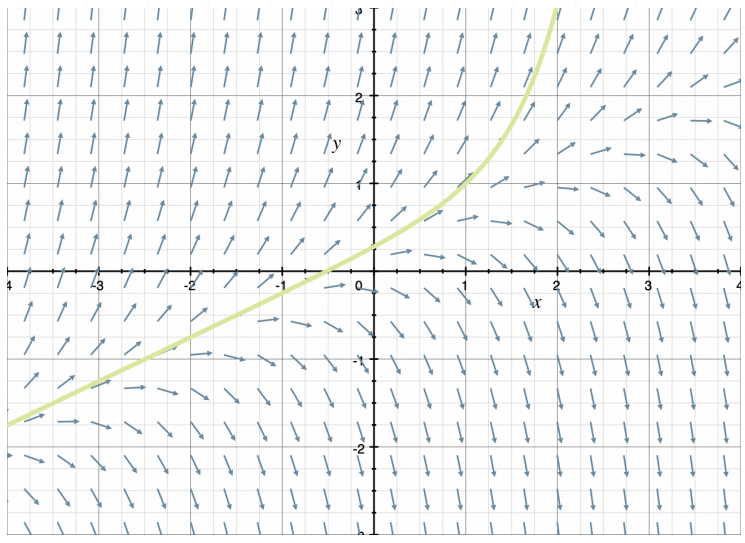
$$y' = -2y + x$$

Answer choices:

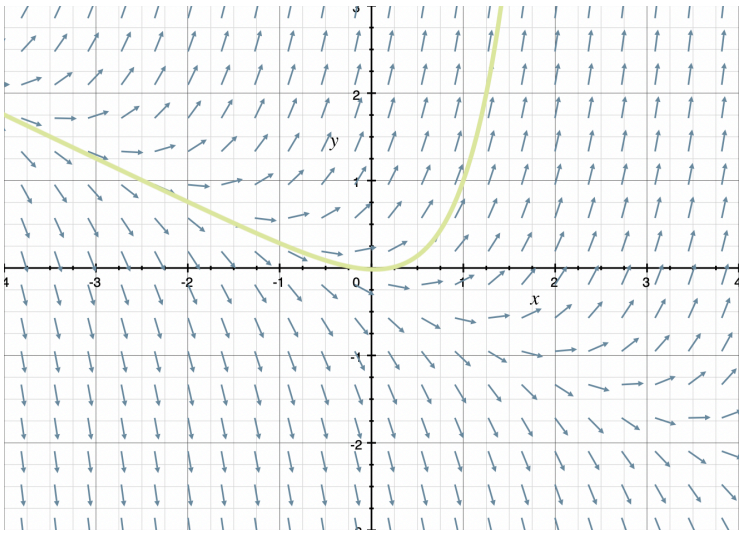
A



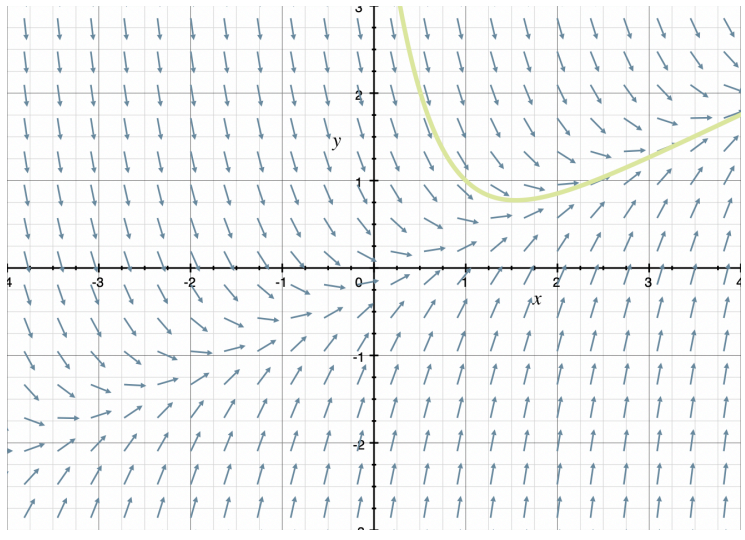
B



C



D



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Solution: D

We'll pair up values $x = \{-2, -1, 0, 1, 2\}$ with values $y = \{-2, -1, 0, 1, 2\}$, calculating y' for each combination of values.

For instance, at $(x, y) = (-2, -2)$, we find

$$y' = -2y + x$$

$$y' = -2(-2) - 2$$

$$y' = 2$$

Or at $(x, y) = (1, 2)$, we find

$$y' = -2y + x$$

$$y' = -2(2) + 1$$

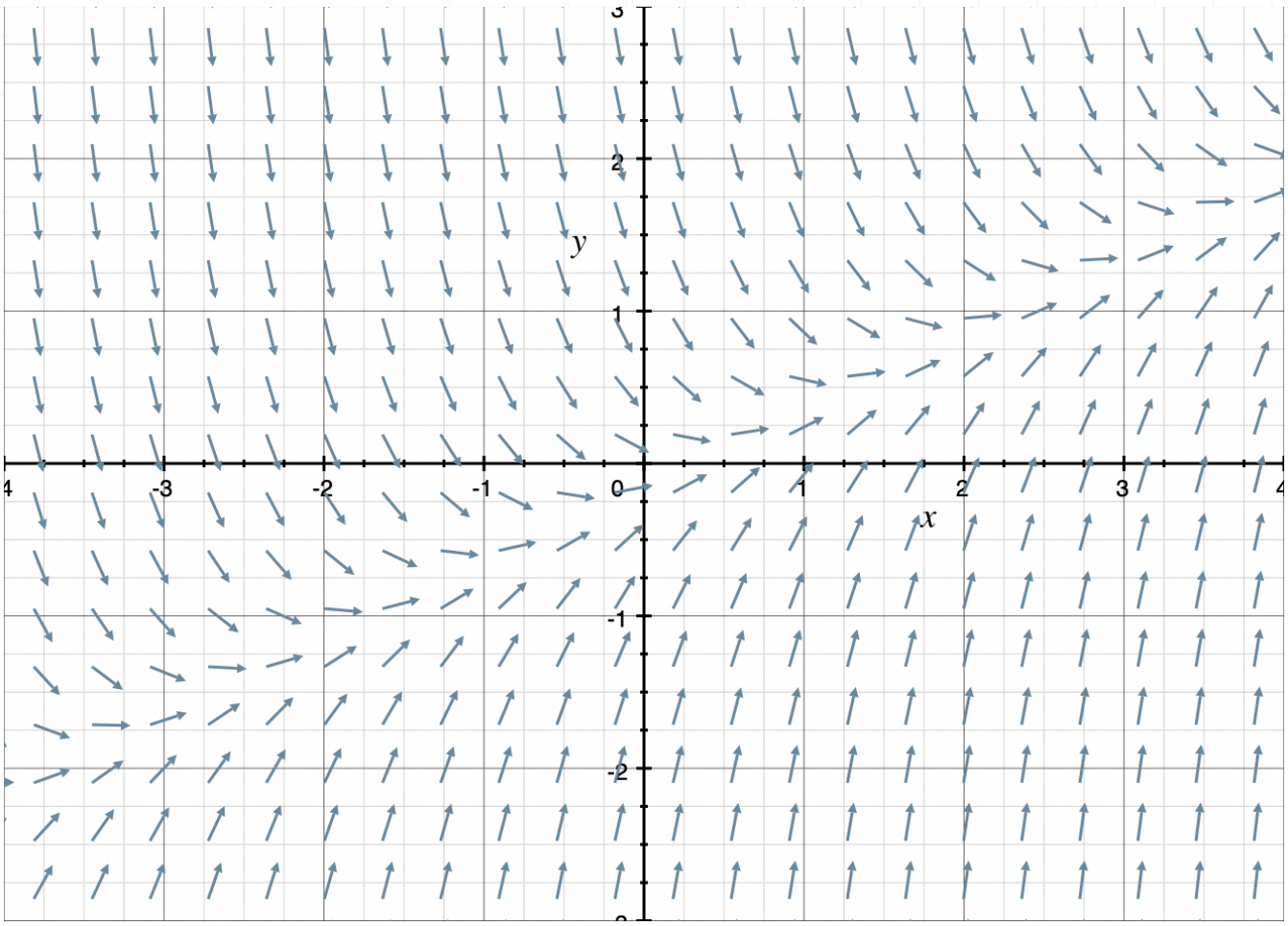
$$y' = -3$$

Let's make a table with each of these combinations.

		x				
		-2	**-1**	**0**	**1**	**2**
y	**-2**	2	3	4	5	6
	-1	0	1	2	3	4
	0	-2	-1	0	1	2
	1	-4	-3	-2	-1	0
	2	-6	-5	-4	-3	-2

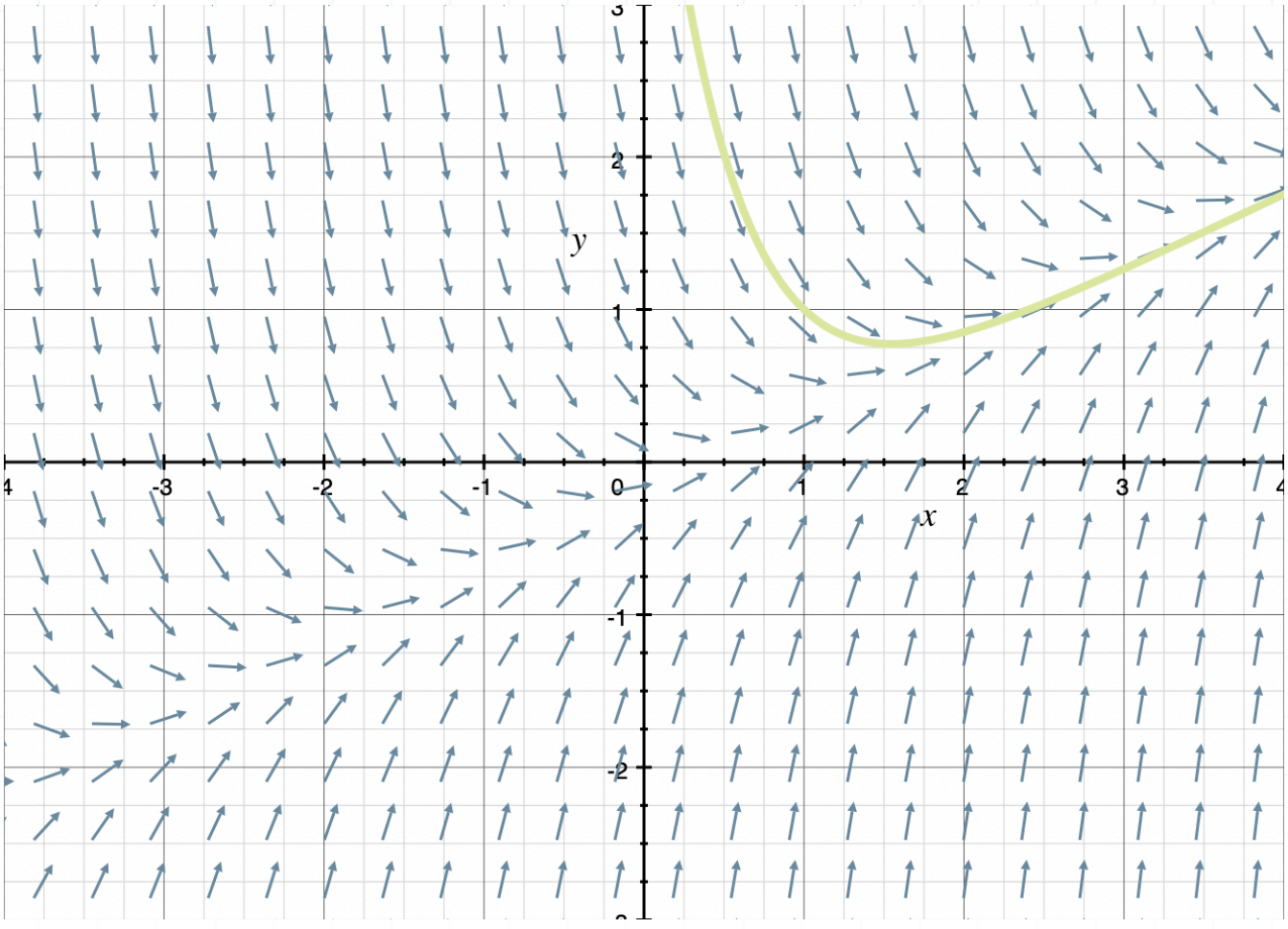
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The values of y' that we found represent the slope of the function at the corresponding point (x, y). For example, we see the point $(2, -2)$, and the corresponding value $y' = 6$. This means that the slope of the function at $(2, -2)$ is 6, so we'd draw a small, short line exactly at $(2, -2)$ with slope 6. Plotting all of the other point-slope pairs, the direction field starts to look something like this:



If we plot the point $(1,1)$, then follow the slopes in our direction field to both the left and right of $(1,1)$, our solution curve should look something like this:

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